

SCIENCE

Science, Grade 12

Workplace Preparation

SNC4E

This course provides students with fundamental science knowledge and workplace skills needed to prepare them for success beyond secondary school. Students will explore hazards in the workplace, chemicals in consumer products, disease and its prevention, electricity at home and at work, and nutritional science. Emphasis is placed on current topics in science and relevant, practical activities that develop students' literacy and mathematical literacy skills and enhance their scientific literacy.

Prerequisite: Science, Grade 10, Applied, or a Grade 10 locally developed compulsory credit (LDCC) course in science

Big Ideas

Hazards in the Workplace

- Knowledge and understanding of science enable people to identify, explain, and minimize hazardous situations in the workplace.

Chemicals in Consumer Products

- Physical and chemical properties of chemicals determine how they can be used in consumer and industrial products.
- Incorrect handling and disposal of chemicals can lead to unsafe conditions in the home and the workplace and can harm the environment.

Disease and Its Prevention

- The spread of communicable diseases can be reduced through personal choices and the use of appropriate technologies.
- Methods used to control the spread of diseases can have both positive and negative effects.

Electricity at Home and Work

- Electrical equipment can pose a safety hazard in the home and workplace if it is not used correctly.
- Rapid advances in electronics, and consumers' desire for the latest electronics technology, have resulted in serious waste disposal problems.

Nutritional Science

- The nutrients and other substances found in food products have effects on human health and well-being.
- Knowledge of the role of nutrients and other substances found in food products enables people to make healthy lifestyle choices.

Fundamental Concepts Covered in This Course (see also page 5)

Fundamental Concepts	Hazards in the Workplace	Chemicals in Consumer Products	Disease and Its Prevention	Electricity at Home and Work	Nutritional Science
Matter		✓	✓		✓
Energy				✓	✓
Systems and Interactions	✓		✓	✓	
Structure and Function			✓	✓	
Sustainability and Stewardship	✓	✓	✓	✓	
Change and Continuity			✓		

A. SCIENTIFIC INVESTIGATION SKILLS AND CAREER EXPLORATION

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1.** demonstrate scientific investigation skills (related to both inquiry and research) in the four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating);
- A2.** identify and describe careers related to the fields of science under study, and describe the contributions of scientists, including Canadians, to those fields.

SPECIFIC EXPECTATIONS

A1. Scientific Investigation Skills

Throughout this course, students will:

Initiating and Planning [IP]*

- A1.1** formulate relevant scientific questions about observed relationships, ideas, problems, or issues, make informed predictions, and/or formulate educated hypotheses to focus inquiries or research
- A1.2** select appropriate instruments (e.g., a decibel meter, spot plates, glassware, thermometers) and materials (e.g., a heat lamp, agar plates, circuit boards), and identify appropriate methods, techniques, and procedures, for each inquiry
- A1.3** identify and locate a variety of print and electronic sources that enable them to address research topics fully and appropriately
- A1.4** apply knowledge and understanding of safe laboratory practices and procedures when planning investigations by correctly interpreting Workplace Hazardous Materials Information System (WHMIS) symbols; by using appropriate techniques for handling and storing laboratory equipment and materials and disposing of laboratory materials; and by using appropriate personal protection

Performing and Recording [PR]*

- A1.5** conduct inquiries, controlling relevant variables, adapting or extending procedures as required, and using appropriate materials and equipment safely, accurately, and effectively, to collect observations and data
- A1.6** compile accurate data from laboratory and other sources, and organize and record the data, using appropriate formats, including tables, flow charts, graphs, and/or diagrams
- A1.7** select, organize, and record relevant information on research topics from a variety of appropriate sources, including electronic, print, and/or human sources, using suitable formats and an accepted form of academic documentation

Analysing and Interpreting [AI]*

- A1.8** synthesize, analyse, interpret, and evaluate qualitative and/or quantitative data to determine whether the evidence supports or refutes the initial prediction or hypothesis and whether it is consistent with scientific theory; identify sources of bias and/or error; and suggest improvements to the inquiry to reduce the likelihood of error
- A1.9** analyse the information gathered from research sources for logic, accuracy, reliability, adequacy, and bias

* The abbreviation(s) for the broad area(s) of investigation skills – IP, PR, AI, and/or C – are provided in square brackets at the end of the expectations in strands B–F to which the particular area(s) relate (see pp. 20–22 for information on scientific investigation skills).

A1.10 draw conclusions based on inquiry results and research findings, and justify their conclusions with reference to scientific knowledge

Communicating [C]*

A1.11 communicate ideas, plans, procedures, results, and conclusions orally, in writing, and/or in electronic presentations, using appropriate language and a variety of formats (e.g., data tables, laboratory reports, presentations, debates, simulations, models)

A1.12 use appropriate numeric, symbolic, and graphic modes of representation, and appropriate units of measurement (e.g., SI and imperial units)

A1.13 express the results of any calculations involving data accurately and precisely, to the appropriate number of decimal places or significant figures

A2. Career Exploration

Throughout this course, students will:

A2.1 identify and describe a variety of careers related to the fields of science under study (e.g., chemical technician, baker, blood laboratory assistant, custodian, public works employee, cosmetologist) and the education and training necessary for these careers

A2.2 describe the contributions of scientists, including Canadians (e.g., Lorne Trottier, David Butler-Jones, Francine Décary, Robert G.E. Murray, Susan Barr), to the fields under study

B. HAZARDS IN THE WORKPLACE*

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** assess common workplace settings with respect to hazards, and analyse selected legislation that is in place to protect workers and the environment from these hazards;
- B2.** investigate the nature of workplace hazards and various ways in which workers can protect themselves from these hazards;
- B3.** demonstrate an understanding of common biological, chemical, and physical workplace hazards.

SPECIFIC EXPECTATIONS

B1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- B1.1** assess a workplace setting, either real or simulated, with respect to hazards that could affect workers or the environment, using appropriate criteria (e.g., a checklist for a health and safety audit) [AI, C]

Sample issue: An employee has been asked to clean out the oil fryer at the restaurant where he works. He knows from his training that the fryer must cool down first, but the manager is rushing the clean-up crew to avoid having to pay them overtime.

Sample questions: What hazardous substances are used in the workplace being assessed? Are warnings posted as to the dangers they pose? How are the substances handled, stored, and disposed of? Are these practices safe? How might they be improved?

- B1.2** analyse and summarize the requirements of selected sections of workplace safety and/or environmental protection legislation related to a career of personal interest (e.g., regulations applying to mining in the Occupational Health and Safety Act; regulations applying to waste management in the Ontario Environmental Protection Act) [AI, C]

Sample issue: Section 91.1 of the Ontario Environmental Protection Act requires employers to prevent or reduce the risk of spills of

pollutants and, if such a spill does occur, to provide the appropriate equipment, personnel, and material to clean it up. This section covers a range of workplaces where spills of environmental contaminants may occur.

Sample questions: What types of jobs are affected by regulations under the Ontario Environmental Protection Act? What types of workers are covered by the Canada Labour Code? What changes did Bill C-45 make to the Canadian Criminal Code? Why? What is the purpose of the Ontario Needle Safety Regulation under the Occupational Health and Safety Act? What measures are in place in Ontario to protect workers from violence and harassment on the job?

B2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- B2.1** use appropriate terminology related to hazards in the workplace, including, but not limited to: *occupational exposure limits (OEL)*, *designated substance regulation (DSR)*, *personal protective equipment (PPE)*, *route of entry*, *controlled product*, *infectious material*, *inhalation*, *absorption*, *ingestion*, *injection*, and *exposure values* [C]
- B2.2** plan and conduct an inquiry to determine what factors affect rates of chemical reactions (e.g., the concentration of an acid affects its rate of reaction with metals) [IP, PR, AI]

* Activities related to the achievement of expectations that refer to “the workplace” and/or that involve hazardous materials may be simulated in the classroom. It is the teacher’s responsibility to ensure students’ safety.

B2.3 investigate the effectiveness of methods used to reduce the impact of noise in the workplace (e.g., use a decibel meter to measure noise level before and after the installation of sound insulation; measure the effectiveness of earplugs at different sound levels) [PR]

B2.4 investigate the effects of workers' exposure to heat or cold (e.g., the effects of industrial heat sources such as molten materials on workers in foundries and factories; the effects of seasonal heat and cold, including exposure to solar radiation, on outdoor workers in construction, landscaping, agriculture, or hydro line repair; the effects of cold on workers in refrigerated warehouses) [PR]

B2.5 use a research process to investigate procedures for the safe handling of biohazardous and/or infectious materials in the workplace, and communicate their findings (e.g., create a webpage on the universal precautions for handling biological hazards; create a poster illustrating the steps for proper hand washing) [IP, PR, C]

B3. Understanding Basic Concepts

By the end of this course, students will:

B3.1 describe the ways in which hazardous materials enter the body (i.e., ingestion, inhalation, absorption, and injection), and explain the importance of using personal protective equipment (e.g., gloves, appropriate eye wear, aprons, self-contained breathing apparatus) to avoid contamination

B3.2 identify common physical hazards in the workplace (e.g., hazards posed by noise; cutting tools; electrical power lines; extreme heat and cold), and describe potentially harmful situations and practices (e.g., work at heights on unstable equipment) as well as best safety practices (e.g., properly securing ladders and scaffolding) relating to these hazards

B3.3 identify common biological hazards in the workplace (e.g., bacteria, viruses, fungi), and describe potentially harmful situations and practices (e.g., improper disposal of syringes) as well as best safety practices (e.g., use of PPE such as gloves and masks) relating to these hazards

B3.4 identify common chemical hazards in the workplace (e.g., oxidizers, acid and base solutions), and describe potentially harmful situations and practices (e.g., inadequate venting of fine dust particles in flour mills) as well as best safety practices (e.g., wearing goggles and a self-contained breathing apparatus when working near substances that can irritate the eyes or lungs) relating to these hazards

B3.5 describe ways in which workers can address safety issues in the workplace (e.g., by reporting an unsafe condition to a supervisor; by refusing unsafe work)

B3.6 explain qualitatively how factors such as temperature, concentration, and the size of the opening of a container affect storage and disposal of chemicals in the workplace

C. CHEMICALS IN CONSUMER PRODUCTS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** analyse chemical products used in the home and workplace, and issues related to their safe and environmentally responsible use and disposal;
- C2.** investigate chemical properties of, and chemical reactions used to produce, various consumer products;
- C3.** demonstrate an understanding of chemical reactions, and of properties of chemicals used in common household and workplace products.

SPECIFIC EXPECTATIONS

C1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- C1.1** analyse, on the basis of research, a chemical product used in a particular profession or in the home (e.g., pool chemicals, chlorine bleach, hair dye), and prepare guidelines for safe and responsible use of the product [IP, PR, AI, C]

Sample issue: Bathroom cleaners need to be strong enough to kill germs and remove stains, but such requirements often mean that these products contain chemicals that can burn the skin and irritate the eyes. What do people need to know about the precautions that should be taken when using such products?

Sample questions: What chemicals are in hair dye? How should hair stylists protect themselves and their clients from the harmful effects of these chemicals? What precautions must be taken when using solvents or bleaches? Why?

- C1.2** assess the environmental consequences of improper disposal of chemical products commonly used in the home (e.g., pouring paint down the drain; dumping batteries in garbage destined for landfill sites) [AI, C]

Sample issue: Some batteries contain lithium or cadmium, which are toxic heavy metals. If such batteries are tossed into the regular garbage,

these metals can leach into the soil or run off into water systems. Batteries can also overheat and cause a fire or other kind of chemical reaction.

Sample questions: What happens when pharmaceuticals are poured down the drain? Why should containers of flammable substances such as turpentine or corrosive substances such as drain cleaner not be thrown in the regular garbage?

- C1.3** evaluate the appropriateness of current disposal practices in their home, at school, or in the community, with particular reference to the disposal of chemical waste [AI, C]

Sample issue: Many commonly used chemicals can damage the environment if they are not properly disposed of. Some chemicals are combustible, produce toxic vapours, or are corrosive. Some otherwise safe chemicals can become toxic if combined with another chemical. Safe disposal methods must take the properties of each chemical into account.

Sample questions: What is the proper method for disposing of solvent-soaked rags? What sorts of items are considered to be household hazardous waste (HHW)? Are there HHW depots in your community? What happens to the waste once it arrives at such a depot?

C2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- C2.1** use appropriate terminology relating to chemical reactions and chemical products, including, but not limited to: *synthesis, decomposition, neutralization, polymerization, combustion, single and double displacement, pH, solvent, organic, inorganic, and dilution* [C]
- C2.2** use an inquiry process to determine how various conditions affect a chemical reaction, by altering the conditions under which a reaction occurs (e.g., temperature, length of time, amount of reactants, pH of a solution), observing the effects of the alterations, and comparing the outcome and final product of each reaction (e.g., make borax slime, then alter the proportion of the ingredients and measure the impact on properties of the product) [IP, PR, AI]
- C2.3** prepare dilutions using concentrated solutions, and observe or measure the changes in properties (e.g., pH, colour, viscosity, density) [PR]
- C2.4** safely conduct a chemical reaction in order to produce a common household or consumer product (e.g., taffy, shampoo, toothpaste, nylon, lip balm) [PR]
- C2.5** classify various household products on the pH scale, using pH paper, indicator solutions, and/or a pH meter [PR, AI]

- C2.6** investigate a variety of consumer products within a given category (e.g., shampoo, window cleaner, disinfectant), focusing on products claiming to be environmentally friendly, and analyse them with respect to selected factors (e.g., cost, effectiveness, impact on the environment) [PR, AI, C]

C3. Understanding Basic Concepts

By the end of this course, students will:

- C3.1** describe the types of chemical reactions (e.g., synthesis, single displacement, double displacement, decomposition, combustion, polymerization, neutralization) and the signs of chemical change in each
- C3.2** explain, in qualitative terms, why some chemical substances mix and others do not (e.g., ethanol and vinegar are both polar and therefore miscible)
- C3.3** explain the function of the pH scale and how pH test results are interpreted
- C3.4** identify organic and inorganic compounds commonly used in the home and workplace (*organic*: fats, oils, fuels, common solvents; *inorganic*: acids and bases, mineral solvents, ammonia, baking soda), and compare their properties

D. DISEASE AND ITS PREVENTION

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** evaluate the impact of public policy initiatives and technological advances intended to control the spread of disease, taking into consideration the failure of some people to follow public health regulations or recommendations;
- D2.** investigate the characteristics, growth, and spread of bacteria, and the effects of aseptic techniques and antibiotics;
- D3.** demonstrate an understanding of the causes, symptoms, and modes of transmission of various diseases, and of strategies to prevent the spread of disease.

SPECIFIC EXPECTATIONS

D1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- D1.1** evaluate the effectiveness of a public policy measure or technological advance intended to control the spread of disease (e.g., mandatory immunization, screening for tuberculosis, quarantine) [AI, C]

Sample issue: The federal government established the Canadian Food Inspection Agency to ensure that the food eaten by Canadians is safe. Despite its regulations, food-borne illnesses such as salmonella, listeriosis, and mad cow disease continue to occur.

Sample questions: How effective are flu shots in reducing the incidence and severity of influenza, particularly among vulnerable populations like the elderly? What are some of the groups or organizations that track and inform the public of health risks? What tools and strategies do they use? How effective have they been in protecting human health?

- D1.2** evaluate the impact, current and/or potential, of an individual's choice not to participate in a public health strategy intended to reduce the spread of disease (e.g., a hospital worker who does not follow recommendations regarding hand washing; a worker in a retirement home who does not get a flu shot) [AI, C]

Sample issue: Some parents are deciding not to immunize their children. Unvaccinated children can contract serious diseases such as polio and diphtheria and spread them to other unvaccinated children. The resurgence of such diseases would place a heavy burden on the health care system.

Sample questions: Why have signs showing proper hand-washing techniques been posted in many public restrooms? What can happen if someone in the food service industry ignores these signs? What can happen if a person infected with HIV refuses to use condoms?

- D1.3** analyse, on the basis of research, the advantages and disadvantages of selected technologies used to try to control disease (e.g., the effectiveness of pharmaceuticals at combating disease; the side effects of a variety of drugs) [IP, PR, AI, C]

Sample issue: After World War II, DDT was widely used to kill insects that carried malaria. After researchers found that the pesticide caused abnormalities in some animals and cancer in humans, it was banned in many countries. However, in some countries with a high incidence of malaria, limited use of DDT continues.

Sample questions: What are the advantages and disadvantages of using insecticidal nets to try to control the spread of malaria? Why is the irradiation of food to eliminate food-borne bacteria controversial?

D2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- D2.1** use appropriate terminology related to the prevention of disease, including, but not limited to: *communicable, non-communicable, microorganism, pathogen, disease, epidemiology, vector, immunization record, quarantine, pandemic, vaccine, antiseptic, sterilization, disinfection, and pasteurization* [C]
- D2.2** conduct an investigation, using safe practices and aseptic techniques, to compare the characteristics and growth of different types of non-pathogenic bacteria [PR, AI]
- D2.3** investigate the effects of various drug therapies (e.g., different antibiotic discs) on the growth of bacteria [PR, AI]
- D2.4** use a simulation (e.g., phenolphthalein and sodium hydroxide; a computer simulation) to demonstrate how diseases can spread through a community, and analyse the results [PR, AI]

D3. Understanding Basic Concepts

By the end of this course, students will:

- D3.1** describe modes of transmission of some communicable diseases, including those that are insect-borne (e.g., malaria, encephalitis),

airborne (e.g., influenza, tuberculosis), water-borne (e.g., cholera, poliomyelitis), sexually transmitted (e.g., HIV/AIDS), and food-borne (e.g., mad cow disease, trichinosis, salmonella)

- D3.2** identify the causes and symptoms of various diseases (e.g., AIDS, influenza, salmonella, West Nile virus), and describe measures intended to prevent their spread
- D3.3** describe the reasons for immunization against specific diseases, the function of records of immunization in Ontario, and the importance of maintaining a personal immunization schedule
- D3.4** describe the use of vaccines, antibiotics, antiseptics, and other medical measures, both conventional and alternative, intended to control disease
- D3.5** explain the differences between bacteria and viruses in terms of their size, structure, and reproduction, and the methods used to control their spread
- D3.6** explain the importance of the proper use, storage, and disposal of medications (e.g., the importance of taking the full course of antibiotics, following directions, keeping medications away from children, monitoring side effects, returning expired medication to a pharmacy for disposal)

E. ELECTRICITY AT HOME AND WORK

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** assess electrical hazards in the home and workplace, and the social and environmental impact of electrical technologies;
- E2.** investigate common electrical devices, including their energy transformations and consumption;
- E3.** demonstrate an understanding of electrical circuits, common electrical devices, and safety procedures related to electric systems.

SPECIFIC EXPECTATIONS

E1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- E1.1** assess the social and environmental impact of electrical technologies, including the impact associated with the manufacture and disposal of electronic devices (e.g., the impact of electrical devices used in the health care field, such as pacemakers or respirators; the impact of energy generation needed to power electrical devices and appliances) [AI, C]

Sample issue: Electronics play an important part in our everyday lives. However, disposal of used electronic equipment is a huge problem. Globally, we generate 30 to 50 million tonnes of electronic waste each year. Much of this waste is shipped to developing countries, where it is incinerated or dumped in landfill sites, practices that release toxic chemicals into the air or soil.

Sample questions: How often do you replace an electronic device because something newer, faster, or more powerful has been developed? What do you do with the older devices? What impact have computers had on society?

- E1.2** assess electrical hazards that can be found at home and in the workplace (e.g., electrical outlets close to areas where spills might occur; overloaded circuits), and propose practical courses of action to address the problems [AI, C]

Sample issue: Portable electric tools are convenient and efficient. However, if used improperly or if basic electrical safety precautions, such as proper grounding, are ignored, such tools can constitute a safety hazard.

Sample questions: What factors are important to consider when wiring a circuit in an area near water? What are the dangers of using electric hair dryers or razors near a sink full of water? What factors should be investigated when purchasing a home to ensure it meets the current electrical safety standards?

E2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- E2.1** use appropriate terminology related to electricity, including, but not limited to: *energy, power, kilowatt-hour, potential difference, current, conductor, short circuit, circuit breaker, fuse, and resistance* [C]
- E2.2** draw energy flow diagrams and/or write energy transformation equations that illustrate the energy transformation occurring in household devices, including the production of waste energy (e.g., energy transformations in a digital music player: electrical energy → kinetic energy + sound energy + light + waste heat energy) [C]
- E2.3** build a simple electrical device or circuit (e.g., a loudspeaker, an electric motor, a D-cell, a circuit containing a 40W lightbulb and a dimmer switch), following a clear set of instructions and diagrams, and using appropriate tools safely [PR]
- E2.4** calculate the electrical energy consumption of two similar appliances (e.g., an old and a new refrigerator), using the power ratings that appear on the appliance, and compare the financial and environmental costs (e.g., carbon dioxide emissions) of running the two appliances [AI]

E2.5 analyse changes in household energy consumption over a given time period (e.g., throughout the course of a day; between a week in January and a week in May), and give reasons for the changes [AI, C]

E3. Understanding Basic Concepts

By the end of this course, students will:

E3.1 describe basic electric circuit components, including those that regulate the flow of electricity or are used as safety mechanisms (e.g., switches, bimetallic strips, resistors, ground fault interrupters [GFIs], surge protectors), and explain their layout in an electric circuit

E3.2 describe forms of energy (e.g., electrical, mechanical, sound, light, thermal) and the energy transformations that occur in common electrical devices, including production of waste energy (e.g., heat)

E3.3 identify situations in which direct current (DC) and alternating current (AC) are used (e.g., DC is used in a portable appliance such as a flashlight; AC is used in a household appliance such as a kettle)

E3.4 explain the difference in voltage requirements, and identify some household appliances that require 110 V AC (e.g., microwave oven, blender) and some that require 220 V AC to operate (e.g., conventional oven, clothes dryer)

E3.5 describe safety procedures to be followed when using electric systems at home or at work (e.g., ensuring that tools and appliances are properly grounded; unplugging appliances by pulling the plug, not the cord), and explain how dangerous situations can occur (e.g., an overloaded circuit can overheat and cause a fire; digging through buried electrical cable can cause a severe shock)

F. NUTRITIONAL SCIENCE

OVERALL EXPECTATIONS

By the end of this course, students will:

- F1.** assess the environmental implications of a variety of food choices, and evaluate and propose ways to improve the nutritional content of a menu;
- F2.** investigate nutrients and non-nutrient additives in a variety of foods;
- F3.** demonstrate an understanding of food components and their effects on the human body.

SPECIFIC EXPECTATIONS

F1. Relating Science to Technology, Society, and the Environment

By the end of this course, students will:

- F1.1** assess the environmental implications of food choices available in a variety of situations (e.g., in the school cafeteria, a fast-food restaurant, a supermarket, a local farmers' market, an organic meat shop), and propose ways to minimize the environmental impact of their food choices [AI, C]

Sample issue: Supermarkets commonly sell imported produce, distributed through large warehouses, even when the same types of food are in season locally and are available from local farmers. Importing foods generates greater carbon emissions but may be seen as more efficient if local farmers lack a reliable distribution system.

Sample questions: What is the environmental impact of organic farming compared to traditional farming methods? What are the advantages and disadvantages of buying certified organic foods from a local farmer? What are the environmental costs of purchasing a pizza? Why is the environmental footprint associated with consuming a hamburger different from that associated with eating a veggie burger?

- F1.2** evaluate the nutritional content of a menu (e.g., from the school cafeteria, a fast-food restaurant, a coffee shop, a retirement home, a hospital), and propose ways to improve it, using information from *Eating Well with Canada's Food Guide* or *Eating Well with Canada's Food Guide: First Nations, Inuit, and Métis* [AI, C]

Sample issue: The menus in fast-food restaurants tend to be high in fat, sodium, and sugar. However, there are many options such restaurants might consider, such as using whole grain buns, limiting the amount of salt added to burgers and fries, and providing additional menu choices that include fruits and vegetables.

Sample questions: Is a salad always a healthy choice at a restaurant? Why or why not? What is the nutritional difference between a serving of French fries and a baked potato served with the skin on? What is the nutritional difference between white bread and whole grain bread? What foods are high in protein? How many daily servings of fruits and vegetables are recommended by Canada's Food Guide? Why?

F2. Developing Skills of Investigation and Communication

By the end of this course, students will:

- F2.1** use appropriate vocabulary related to nutritional science, including, but not limited to: *nutrient, lipid, carbohydrate, protein, vitamin, mineral, qualitative test, serving size, food additive, trans fat, cholesterol, kilojoule, calorie, saturated, unsaturated, hydrogenated, essential amino acid, and preservative* [C]
- F2.2** conduct an investigation to compare commercial food products and home-made foods (e.g., commercial and home-made cookies or cake; a shake from a fast-food restaurant and a home-made milk shake; commercial orange

drink and freshly squeezed orange juice), with reference to qualitative and quantitative differences such as the number of ingredients, types of nutrients, non-nutrient additives, texture, and colour [PR, AI]

F2.3 conduct an investigation to test for the presence of various nutrients in foods (e.g., use iodine to test for starch; use Benedict's solution to test for simple sugar) [PR]

F2.4 plan and conduct an investigation into the effectiveness of food preservatives (e.g., use lemon juice to reduce oxidation of apple slices; compare mould growth on commercial and home-made bread) [IP, PR]

F2.5 modify a recipe or menu to meet a dietary restriction (e.g., reduce the cholesterol content by replacing whole eggs with egg whites; reduce the sodium content by cutting salt; replace milk with soy milk; replace meat with tofu or legumes), and explain the reasons for the changes (e.g., sodium can contribute to high blood pressure; dairy products can cause digestive problems for people who are lactose intolerant; non-animal sources of protein are necessary for vegans, who do not eat any animal products) [PR, C]

F3. Understanding Basic Concepts

By the end of this course, students will:

F3.1 identify sources of the principal food nutrients (e.g., carbohydrates, lipids, proteins, vitamins, minerals, fibre), with reference to Canada's Food Guide, and describe the function of these nutrients in the body

F3.2 identify the type of information commonly found on a food label, and describe how the information is organized (e.g., serving size, percentage of daily values, amount of each component)

F3.3 explain the meaning of a variety of descriptors found on food labels (e.g., "fat free", "low fat", "lite", "pure", "organic", "lean", "diet")

F3.4 describe the function of non-nutrient food additives (e.g., lecithin; monosodium glutamate [MSG]; artificial colour, flavour, and sweetener; preservatives), and explain their effects on human health